

MTH204 (Section A)
Graded Worksheet–6

Name:

Roll No.:

Important Notice: The use of electronic gadgets is strictly prohibited during the tutorial. This includes mobile phones, tablets, or any device containing class notes, but anything handwritten is completely allowed.

Q–1) Determine the location, classification and stability of all critical points of system of ODEs:

$$y_1' = y_2, \quad y_2' = -y_1 + \frac{1}{2}y_1^2.$$

Q–2) Find a general solution of the following system using variation of parameters:

$$y_1' = y_2 + e^{3t}, \quad y_2' = y_1 - 3e^{3t}.$$

Q–3) Find a general solution of the following system using method of undetermined coefficients:

$$y_1' = 4y_1 + y_2 + 0.6t, \quad y_2' = 2y_1 + 3y_2 - 2.5t.$$

Q–4) Find the location and type of all critical points by first converting the ODE to a system and then linearizing it:

$$y'' - 9y + y^3 = 0.$$

Solution:

Solution:

Solution: